14. Construct a C Program to Organize the File Using Single Level Directory

Code - Part 1 (Lines 1–36) — Copy Function

```c
1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <fcntl.h>
4  #include <unistd.h>
5
6  #define BUFFER_SIZE 4096
7
8  void copy()
9  {
10     const char *sourcefile = "sasi.txt";
11     const char *destination_file = "sk.txt";
12
13     int source_fd = open(sourcefile, O_RDONLY);
14     if (source_fd < 0)
15     {
16         printf("Error opening source file\n");
17         return;
18     }
19
20     int dest_fd = open(destination_file, O_WRONLY | O_CREAT | O_TRUNC, 0666);
21     if (dest_fd < 0)
22     {
23         printf("Error creating destination file\n");
24         close(source_fd);
25         return;
26     }
27
28     char buffer[BUFFER_SIZE];
29     ssize_t bytesRead, bytesWritten;
30
31     while ((bytesRead = read(source_fd, buffer, BUFFER_SIZE)) > 0)
32     {
33         bytesWritten = write(dest_fd, buffer, bytesRead);
34     }
35
36     close(source_fd);
```

Code - Part 2 (Lines 36–66) — Create & Delete Functions

```
36      close(source_fd);
37      close(dest_fd);
38
39      printf("File copied successfully.\n");
40  }
41
42  void create()
43  {
44      FILE *fp = fopen("sasi.txt", "w");
45      if (fp == NULL)
46      {
47          printf("Error creating file\n");
48          return;
49      }
50
51      printf("File created successfully.\n");
52      fclose(fp);
53  }
54
55  void deleteFile()
56  {
57      if (remove("sasi.txt") == 0)
58          printf("File deleted successfully.\n");
59      else
60          printf("Error deleting file.\n");
61  }
62
63  int main()
64  {
65      int n;
66
```

Code - Part 3 (Lines 67–86) — Main Function

```
67      printf("1. Create\n2. Copy\n3. Delete\nEnter your choice: ");
68      scanf("%d", &n);
69
70      switch(n)
71      {
72          case 1:
73              create();
74              break;
75          case 2:
76              copy();
77              break;
78          case 3:
79              deleteFile();
80              break;
81          default:
82              printf("Invalid choice\n");
83      }
84
85      return 0;
86  }
```

Output

```
1. Create
2. Copy
3. Delete
Enter your choice: 2
Error opening source file


...Program finished with exit code 0
Press ENTER to exit console.
```